CHAPTER 2

ENTERPRISE SYSTEM DEVELOPMENT AND DEPLOYMENT

Quotation

Enterprise Systems include applications and associated information and have as a fundamental premise a structure that permits the integration of data held in any enterprise system with that held in other enterprise systems through the use of appropriate common identifiers.

Structure

- *Quick look at chapter themes*
- *Learning objectives*
- *Case study:* **The State of South Carolina EIS**
- *Sketch of EIS development*
- *Introduction*
  - 2.1 The Manufacturing Roots of ERP
  - 2.2 Stages
  - 2.3 Historical Background
  - 2.4 Evolution of ERP
  - 2.5 Material Requirements Planning (MRP-I)
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  - 2.8 Comparative Coverage Between MRP, ERP, EIS
  - 2.9 EIS as per the Gartner View

2.10 Conclusion
☆ Keywords
☆ Terminology Review
☆ Competency Review
☆ Suggested Reading

♣ QUICK LOOK AT CHAPTER THEMES

Although most people have a dissimilar outlook about 'the new economy' everyone seems to agree about at least one thing, i.e. the Internet which is going to radically transform how companies do business. But the million dollar questions are: how is this expansion in reality going to take place? Will the economy as we are acquainted with it change? Manufacturing software options used to be much simpler than the choices that confront manufacturers today. In the old days quality and price were the primary concerns of the buyer. As additional products became available, convenience was added to the mix. If a local manufacturer had good quality and a reasonable price, that company could get the sale over some distant company who might have a better price. Enterprise system provides a tightly integrated solution to an organization's information system needs for both manufacturing and service organizations. In the last decade, nomenclature has evolved from material requirements planning to manufacturing resource planning to enterprise resource planning [ERP] and now enterprise information system, as added functionality has been added.

♣ LEARNING OBJECTIVES

In highly competitive business, firms repeatedly have to review their business objectives to accomplish their organizational purposes. EIS technologies provide the means to incorporate strategic business solutions within and across the component parts of organizational information system infrastructures. The learning objective of this chapter is to introduce you to the world of enterprise systems and enable you to:

- ♣ Understand manufacturing operating systems which determine how you make your product, maintain quality, supervise operations, purchase components and materials, and handle customer orders and complaints, and other activities.
- ♣ Guesstimate the costs and benefits obtained from MRP-I and MRP-II implementation.
- ♣ Understand the set-up of manufacturing systems including processes, quality assurance, production control, purchasing and customer service.
- ♣ Understand how manufacturing management systems have developed as part of MRP (material requirement planning).
- ♣ Desire to standardize and improve business processes.
- ♣ Aim to assimilate the organization's existing information systems.
- ♣ Understand the need for enhanced and timelier information.
- ♣ Review development of Enterprise systems.
- ♣ Present the concept of Enterprise systems, specially the ERP modules.
- ♣ View relative use and modification of modules.

♣ CASE STUDY

The State of South Carolina EIS

The State of South Carolina has undertaken a statewide effort to transform the way state government operates. The state is putting into place enterprise systems to support efforts to better gauge the price of government. South Carolina has implemented a pilot project and a business case study for their current effort to deliver an enterprise solution that will among other things: 1. Reduce the time and cost to provide quality customer service. 2. Provide tools for monitoring and measuring cost savings and process improvements estimated at $100M per year. 3. Deliver enabling technology to allow for use by persons with disabilities. South Carolina's goal is to use this enterprise information system to automatically collect information on programs and activities, determine the actual cost of doing business, align resources, and establish consistent business processes and to better serve their constituents.

APPLICATION

Pilot Implementation

The objective of this large pilot project included a sweeping reengineering of out-of-date processes as well as improving information available to management and others. Processes had become burdensome and costly. For example, on average the State spends more than $22 to process a single vendor invoice for payment. Yet, the benchmark for best practice is $10 apiece — and some organizations have cut this cost to less than $5 per invoice. The Comptroller General's Office and the pilot agency also wanted an opportunity to gain valuable experience with the software and its implementation, with a view toward implementation as a State enterprise system. These objectives necessitated a fully integrated system that would provide access to current information; eliminate unnecessary paperwork, reviews, approvals, and duplication of efforts; adopt best practices; significantly reduce manual processes; implement document management for electronic routing and approvals of documents; and interface new systems with the Comptroller General's legacy systems.

Business Case Study

To fully evaluate the merits of the project, the State chose to invest in a return-on-investment (ROI) analysis and prepared a formal business case study to document the expected costs and savings from an enterprise implementation. The team conducted an extensive assessment of the State's cost of doing business today, to provide an accurate baseline for comparison. The team then determined the potential savings and benefits available to the State by implementing the solution beyond the pilot agency. The business case study identified potential annual savings of more than $120 million annually by adopting best practices and reengineering its business processes accordingly. These potential savings were centered on 4 major areas: 1. Finance and General Accounting. 2. Procurement and Inventory Management. 3. Human Resources and Payroll. 4. Document Management. When compared to the costs of implementation, if only 10 percent–20 percent of these potential savings are realized, the project pays for itself in 5–7 years.

BENEFITS

Pilot Implementation

By taking an enterprise approach to running government, the State will improve government's bottom line — service delivery. This is not an idealistic dream, but a pragmatic reality. The agency is now using enterprise-wide systems to automatically collect information on programs and activities, determine the actual cost of doing business, align resources and results, establish consistent business processes, and better serve its customers. Actual annual process savings of approximately $600,000 have been documented by our pilot agency. This was despite the fact that the pilot agency continues to have to work within the limits of the State's current inefficient business processes. With this experience, the State now has a basis and the tools to expand on this success.

Business Case Study

The business case study first gives the State objective information on which to base the feasibility of implementing the project. In addition it will also give the state a yardstick by which to measure the ultimate success of an enterprise solution.

IMPORTANCE

Information technology plays a key-enabling role for the State to reengineer its currently inefficient business processes. Without rock solid technology provided by the SAP software, any effort to implement an enterprise solution would fail.

ORIGINALITY

While ERP technology is not ground breaking, governments have been slow to adopt it. As such, enterprise implementation is somewhat leading edge for state governments. That is certainly the case in South Carolina. Additionally, the use of a business case study to justify and move a project forward while not ground-breaking, has not been done to this level before.

SUCCESS

Pilot Implementation

The pilot project was a success in many ways. First it was completed on time and within budget. It resulted in documented savings. To illustrate, prior to implementation, the central procurement function of the agency averaged a backlog of approximately 1 week. After implementation, this backlog was eliminated. In fact central buyers can now concentrate on being true buyers, rather that paper shuffling clerks. In addition the pilot project allowed us a real world example from which to learn valuable lessons on how to approach the enterprise implementation. These lessons learned will be invaluable to the project team as we move forward through the enterprise implementation.

DIFFICULTY

The technology involved, while complex, is certainly not the largest challenge for the project. The cultural aspects of the business process reengineering represent the highest hurdle to clear. People who have been following what are essentially the same business practices for over 20 years are sometimes very slow to accept change. The change management piece of this cannot be under-estimated. This includes accurate and consistent communications along with training and risk management.

Source: http://www.cwhonors.org/laureates/government/20055400.pdf

♣ SKETCH OF EIS DEVELOPMENT

EIS software ideally covers all areas in an organization cutting across functional boundaries and business procedures. An EIS covers a broad array of automated tasks such as SCM (Supply Chain Management), Inventory Replenishment, CRM (Customer Relationship Management), MRP-I (Material Requirement Planning), and MRP-II (Manufacturing Resource Planning). What were stand-alone applications like inventory sales and receivables, financial etc. are integrated in an EIS. An EIS covers all business dimensions of the company such as people, method, structures and technology. Thus, the unique nature of EIS is that it covers all mutually dependent functions in day-to-day operations of an organization. An expert in the field of information system has been quoted as saying that, "It is important that companies know exactly what ERP can do; it is the building blocks on which you add other elements that give you competitive advantage." Thus ERP/EIS is about connecting business systems with computer systems including human beings. Moreover, an ERP/EIS system can be used as a tool to improve the performance level of a supply chain network by helping to reduce cycle times.[1]

♣ INTRODUCTION

Enterprise information systems can be traced back to inventory control packages, material requirements planning (MRP-I), and manufacturing resource planning (MRP II). An inventory control system was software performing conventional inventory processes. It was one of the early

[1] Gardiner, S. C., Hanna, J. B. and LaTour, M. S. (2002). ERP and the re-engineering of industrial marketing processes: A prescriptive overview for the new-age marketing manager. *Industrial Marketing Management*, 31, 357–365.

business applications, which was not integrated with finance and accounting. ERP systems evolved from 1970s-era systems that helped manufacturers identify component and material requirements and availability, leading to a comprehensive and integrated set of functionalities. Now, they have become the organization's information keystone integrating a wide variety of functions including sales and distribution, planning, purchasing, production, cost accounting and finance. Some systems also include engineering management, finite scheduling, production control, and enterprise information system capabilities. The power and benefit of ERP/EIS systems lie in their ability to "mirror" or "model," in electronic form, the company's core and supportive business processes. They provide a global view of an organization, help better manage and utilize company resources, while minimizing duplication and potential for errors. EIS accomplish this objective by modeling business processes. For example, an EIS provides the capability to track a sales order from the order desk, to production planning, purchasing, production, warehousing, shipping and accounts receivable (referred to as "Order to Payment" process). An EIS will also help identify the impact of the order on other departments, for example, purchasing and production, to determine additional material and capacity requirements to fulfill orders.

In today's competitive business environment, companies try to provide customers with goods and services faster and cheaper than their competition. How do they do that? Often, the key is to have efficient, integrated information systems. Increasing the efficiency of information systems results in more efficient management of business processes. When companies have efficient business processes, they can be more competitive in the marketplace.

2.1 THE MANUFACTURING ROOTS OF ERP

The term "MRP" is used as a general term to include all MRP versions, namely, MRP I (i.e. material requirements planning), Closed-loop MRP

(i.e. MRP I with capacity planning and shop floor), and MRP II (i.e. Closed-loop MRP integrated with the other functions such as finance and marketing).[2,3] The concept of an integrated information system took shape on the factory floor. Manufacturing software developed during the 1960s and 1970s, evolving from simple inventory tracking systems to material requirements planning (MRP) software. MRP at its core is a time-phased order release system that schedules and releases manufacturing work orders and purchase orders, so that subassemblies and components arrive at the assembly station just as they are required. Some of the benefits of MRP are reduction of inventories, improved customer service, enhanced efficiency and effectiveness.[4] MRP software allowed a plant manager to plan production and raw materials requirements by working backwards from the sales forecast, the prediction of future sales. Thus, the manager first looked at Marketing and Sales' forecast of demand (what the customer wants); looked at the Production schedule needed to meet that demand; calculated the raw materials needed to meet production; and finally projected raw materials purchase orders to suppliers. For a company with many products, raw materials, and shared production resources, this kind of projection was impossible without a computer to keep track of various inputs.

The basic functions of MRP could be handled by mainframe computers using sequential file processing. Electronic data Interchange (EDI), the direct computer-to-computer exchange of standard business documents, allowed companies to handle the purchasing process electronically, avoiding the cost and delays resulting from paper purchase order and invoice systems. The functional area now known

[2] Duchessi, P., Schaninger, C., Hobbs, D. and Pentak, L. (1988). Determinants of success in implementing material requirements planning (MRP). *Manufacturing and Operations Management*, 1, 263–304.

[3] Sum, C., Yang, K., Ang, J. and Quek, S. (1995). An analysis of material requirements planning (MRP) benefits using alternating conditional expectation (ACE). *Operations Management*, 13, 35–58.

[4] Siriginidi, S. R. (2000). Enterprise resource planning in re-engineering business. *Business Process Management Journal*, 6(5), 376–391.

as Supply Chain Management (SCM) began with the sharing of long-range production schedules between manufacturers and their suppliers.

2.2 STAGES

Companies realized that the management and flow of information was just as important as materials and inventory management. ERP systems are rooted in Material Requirements Planning (MRP) and its Manufacturing Resource Planning (MRP-II) System.[5] ERP has also evolved considerably with computer and technology advances.

There are various stages to ERP (Figure 2.1), which can be explained as follows:

- MRP-I: Material requirements planning which was the historical background of ERP; its purpose being to tap inventory i.e. raw materials planning etc.

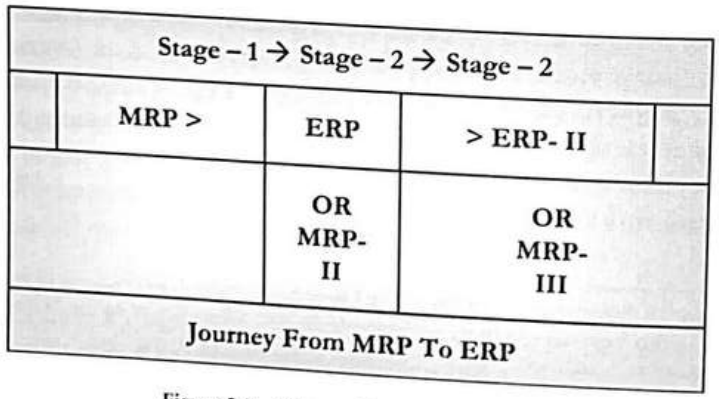


Figure 2.1: Stages of Enterprise Systems.

[5] Hong, K. K. and Kim, Y.-G. (2002). The critical success factors for ERP implementation: An organizational fit perspective. *Information & Management*, 40(1), 25–40.

- MRP-II: Manufacturing resource planning which looks after production-related activities. The concept of MRP-II evolved to look after the shop floor and distribution management activities.
- ERP: Enterprise resource planning has a broader role and is not confined to one department but has an elaborate purview.
- ERP-II or MRP-III: Money resource planning or ERP-II emphasises more on planning of capital or managing the situation when surplus money arises.

  - 1960s The focus of manufacturing systems in this era was on Inventory Control. Software packages were designed to handle inventory based on traditional inventory concepts.
  - 1970s MRP (Material Requirements Planning) systems emerged in this era. This system translated the Master Schedule built for the end items into time-phased net requirements for the sub-assemblies, components, and raw materials planning and procurement.
  - 1980s The concept of MRP I evolved to extend MRP to the shop floor and distribution management activities.
  - 1990s The term MRP I was extended in to a new form known as ERP which covers areas like engineering, finance, human resources, and project management as a whole. Enterprise resource planning is a technological approach for EIS.
  - 2000s ERP-II or MRP III (Money Resource Planning) originated with the motive to emphasize on the planning of money in a optimal manner.
  - 2005s The term EIS was formed with the objective of providing a contemporary version which included web enabled features, full integration, multi-enabled (Figure 2.2).

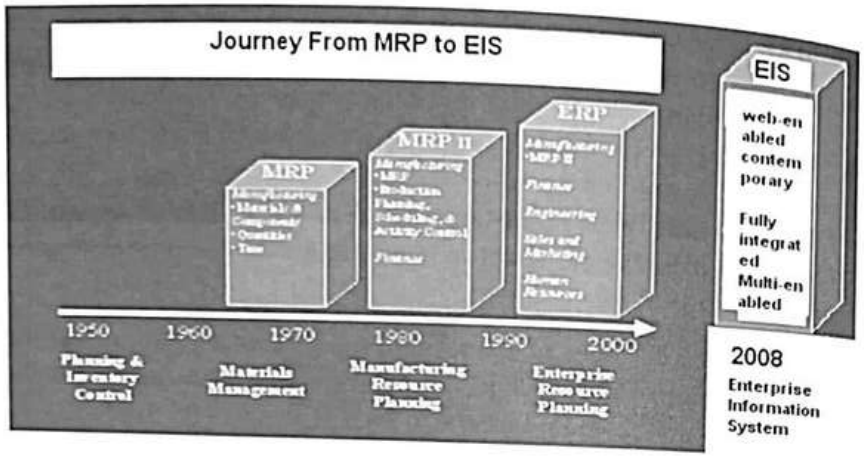


Figure 2.2: EIS Development.

2.3 HISTORICAL BACKGROUND

ERP continues to be one of the largest, fastest-growing and most influential players in the application software industry.[6] ERP (Enterprise Resource Planning) is an industry term for the broad set of activities supported by multi-module application software that helps an organization in the important parts of its business, including product planning, parts purchasing, maintaining inventories, interacting with suppliers, customer service, and tracking orders. ERP can also include application modules for the finance and human resources aspects of a business. Characteristically, an ERP system uses or is integrated with a relational database system. The deployment of an ERP system can involve considerable business process analysis, employee retraining, and new work procedures.

Information available shows that 70 percent of Fortune 1000 firms have or will soon install ERP systems, which will boost the global ERP market from $15 billion already installed as of now to $50 billion over the

[6] Adam, F. and O'Doherty, P. (2000). Lessons from enterprise resource-planning implementations in Ireland: Towards smaller and shorter ERP projects. *Journal of Information Technology*, 15, 305–316.

next five years. ERP applications make up the largest segment of Information Technology budgets. Thirty-nine percent of large companies and 60 percent of smaller companies are deploying ERP systems. The expected annual growth rate for the ERP market over the next five years is 37 percent. With the power ERP exercises over business, it is imperative to see who the key vendors of ERP software are.

ERP integrates the old isolated computer systems in Finance, Human Resource, Manufacturing and Warehousing, and replaces them with a single unified software program organized into software modules. Finance, Manufacturing and Inventory all still get their own software, except that now the software is linked together so that someone in finance can look into the warehouse software to see if an order has been shipped. Most vendors' ERP software is flexible enough so that you can install some modules without buying the whole package. Many companies, for example, will immediately install an ERP finance or HR module and leave the rest of the system for later.

Enterprise resource planning systems arose from a variety of origins. SAP developed their product around supporting the function of manufacturing and integrating that with financial and accounting functions. Other vendors developed their products from other sources. For instance, PeopleSoft began by developing a respected human resources software product, which they expanded to include a slate of other modules. Prior to entry into the ERP market directly, Oracle was the leading database software vendor.

Development of ERP

In the early 1970s, business computing relied upon centralized mainframe computer systems. These systems proved their value by providing a systematic way to measure what businesses did financially. The reports these systems delivered could be used for analysis of variance with budgets and plans, and served as a place to archive business data. Computing provided a way to keep records much more accurately, and on a massively larger scale than was possible through manual means. But from our perspective at the beginning of the 21st century, that level of computer support was primitive.

Business computing systems were initially applied to those functions that were easiest to automate, and that called for the greatest levels of

consistency and accuracy. Payroll and accounting functions were an obvious initial application. Computers can be programmed to generate accurate paychecks, considering tax and overtime regulations of any degree of complexity. They also can implement accounting systems for tax, cost, and other purposes; because these functional applications tend to have precise rules that cover almost every case, computers can be entrusted to automatically and rapidly take care of everything related to these functions.

Prior to 2000, ERP systems catered to very large firms which could afford the rather high costs of purchasing ERP systems. Even focusing on a selected few modules would typically cost firms $5 million and up for software. After 2000, demand dropped, in part because firms were often concerned with Y2K issues prior to 2000 which motivated many ERP system acquisitions. Demand noticeably dropped off after 2000 came and went. Vendors reacted in a number of ways. First, the market consolidated, with Oracle purchasing PeopleSoft (which had earlier acquired JD Edwards). Microsoft acquired a number of smaller ERP software products, consolidating them into Microsoft Dynamics which caters to a smaller priced market, thus serving a needed gap in ERP coverage for small businesses. Notably, SAP advertises that they can serve small business too. But it appears that they are more valuable in the large-scale enterprise market. There are also many other systems to include open sourced systems (at least for acquisition) like Compiere in France. Many countries, such as China, India, and others have thriving markets for ERP systems designed specifically for local conditions, although SAP and Oracle have customers all over the globe.

2.4 EVOLUTION OF ERP

In an ever-growing business environment the following demands are placed on organizations:

- Emphasis on customers rather than other factors like market share, brand image etc.
- Aggressive cost control initiatives.

- Need to analyze costs/revenues by product or customer.
- Flexibility to respond to changing business requirements.
- More informed management decision-making.
- Changes in ways of doing business.

Difficulties in getting accurate data, timely information and improper interfaces across business functions have been identified as hurdles in the growth of any business. Depending upon the velocity of the growing business needs, various applications and planning systems have been introduced into the business world for overcoming these obstacles and to achieve required growth. They include:

- Management Information Systems (MIS).
- Integrated Information Systems (IIS).
- Artificial Intelligence (AI).
- Corporate Information Systems (CIS).
- Knowledge Management (KM).
- Enterprise Wide Systems (EWS).
- Business Intelligence System (BIS).
- Material Requirement Planning (MRP).
- Manufacturing Resource Planning (MRP-II).
- Money Resource Planning (MRP-III).

The latest planning tool added to the above list is Enterprise Resource Planning. Any ERP implementation is a special event since it involves the entire organization with long-term impact. It brings together different functionalities, people, procedures and philosophies, and brings sweeping changes throughout the enterprise. ERP implementation is one of the top growing segments in the information technology industry today. To take advantage of emerging technologies and business practices and meet the evolving business requirements of a thriving industry, companies like People Soft, SAP-AG, BaaN, QaD, IFS, Siebel, Oracle, JD-Edward, Makess, Ramco, etc. have launched their products in this field. Shrinking geographical borders, integration of currencies, ever decreasing product life cycles, condensed profit margins and the need to raise productivity can all be addressed by ERP solutions.

2.5 MATERIAL REQUIREMENTS PLANNING (MRP-I)

The MRP system begins with three documents. A forecast for end-items being assembled is needed by time period. A bill of materials (BOM) describes the components that go into an assembled product. It lists each part in a hierarchical tree, by quantities required for each subassembly, all the way up to the final end-item. Finally, inventory records describing quantities of each component on hand is needed, as well as ordering policies with vendors and lead times. Relationships of these and other MRP documents are given in Figure 2.3.

The material requirements planning document should provide four basic items of information, when to place the order, how much to order, who to order from and when the items need to be on hand.[7] Material requirements planning systems are used to acquire or fabricate component quantities on time both for internal purposes and/or sales and distribution. MRP is a planning instrument that is geared exclusively to assembly

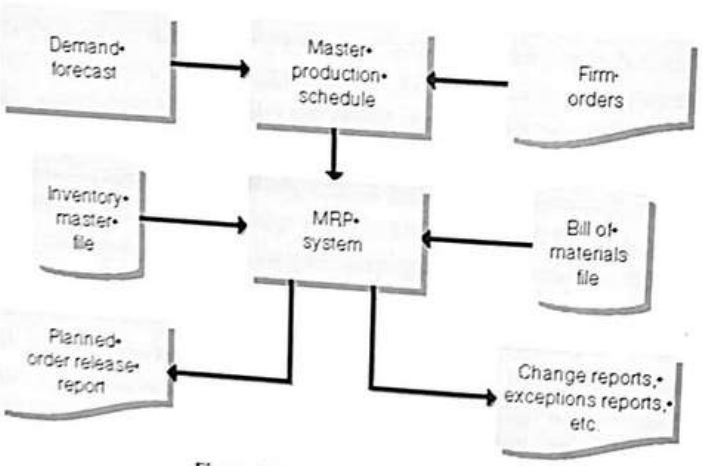


Figure 2.3: MRP Documents.
Source: John Wiley & Sons, Inc. 2006.

[7] www.invatol.com, 2008.

operations. Each manufacturing unit informs its suppliers what parts it needs and when it requires them. The main aim for evolving MRP was to tackle the problem of "dependent demand"; determining how several of a particular component is required to produce the number of finished products.

MRP-I features include:

- Focus on time delivery.
- Incorporate in-house and vendor lead times.
- The date the order is required can be used as the starting date from which to develop the schedule.
- Lead times are known, or can be estimated, in advance.
- MRP calculates procurement quantities and dates as well as plans the corresponding procurement elements.
- MRP calculations are time-phased, i.e. material should be in stock following the JIT philosophy, not earlier, not later, so it is extremely important to get accurate lead times.
- Reduce inventory levels.
- Requirements sorted by vendor.
- Specifying the type, quantity, and time of requirements, in addition to calculating when and for what quantity an order has to be created.
- Time-phased orders.
- Value-projected inventory.
- Valued-projected inventory balance report.

Figure 2.3 shows the hierarchy of relationships of these systems.

Most manufacturing problems would be a lot easier if you could merely perceive them coming. Materials planning entails managing sales forecasts, creating master schedules, and running MRP. In layman's terms, this is equalizing future supply and demand. MRP enables one to plan for the future. Thus MRP involves the monitoring of stocks and, in particular, the mechanical formation of procurement proposals for purchasing and production. SAP's view of MRP is given in Figure 2.4, while Table 2.1 displays input and output information.

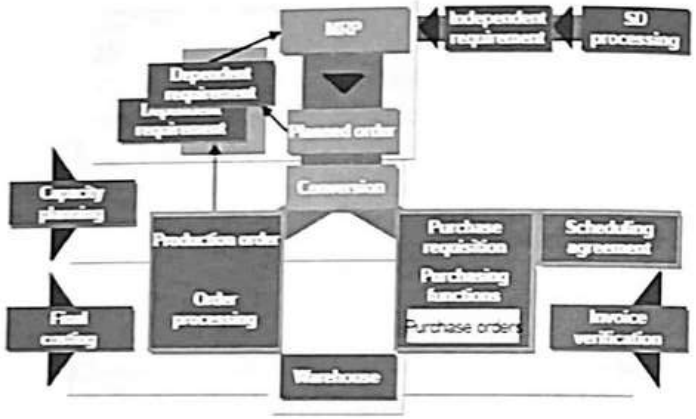


Figure 2.4: SAP View of MRP Documents.

Source: SAP AG Germany, 2008.

Table 2.1: Information Needed and Obtained in MRP-I.

Information needed for MRP-I	Information obtained from MRP-I
♣ Demand for all products. ♣ Lead times for all finished goods, components, parts and raw materials. ♣ Lot-sizing policies for all parts. ♣ Opening inventory levels. ♣ Safety stock requirements. ♣ Any orders previously placed but not arrived yet.	♣ Planned orders: replenishment orders to be released at a future time. ♣ Order release notice: notices to release planned orders. ♣ Action notices: notices to expedite, de-expedite, or cancel orders, or to change order quantities or due dates. ♣ Priority reports: information regarding which orders should be given priority. ♣ Inventory status information. ♣ Performance reports such as inactive items, actual lead times, late orders, etc.

Source: http://www.uoguelph.ca/~dsparlin/mrp.htm, 2008.

Module MM covers the functions of MRP. MRP began as an inventory re-ordering tool in operations involving dependent demand (the demand for materials depends upon the demand for end-items in which the materials are used). The capability of MRP systems evolved to support planning

of all company resources, and currently can support business planning, production planning, purchasing, inventory control, shop floor control, cost management, capacity planning, and logistics management. The use of MRP resulted in better inventory and raw materials control, reduced need for clerical support, and reduced lead times in obtaining materials. Improved communication and better integration of planning were also gained.

2.6 MANUFACTURING RESOURCE PLANNING (MRP-II)

Like MRP-I, MRP-II focuses on the manufacturing process. The next stage of MRP-II evolution was just-in-time (JIT) methodology in the late 1980s. **MRP-II (manufacturing resource planning)** is a method to plan all resources for a manufacturer. A variety of business functions are tied into MRP-II systems to include order processing as in MRP, business planning, sales and operations planning, production plans, master production scheduling, capacity requirements planning, and capacity planning. MRP-II systems are integrated with accounting and finance subsystems to produce reports including business plans, shipping budgets, inventory projections, and purchase plans. A major purpose of MRP-II is to integrate primary functions (i.e. production, marketing and finance) and other functions such as personnel, engineering and purchasing into the planning process to improve the efficiency of the manufacturing enterprise.[8,9]

There is a tendency within the operations management field to consider ERP as a natural extension of MRP-II. Manetti (2001)[10] gave the

[8] Chung, S. H. and Snyder, C. A. (2000). ERP adoption: A technological evolution approach. *International Journal of Agile Management Systems*, 2(1), 24–32.

[9] Mabert, V. A., Soni, A. and Venkataramanan, M. A. (2001). Enterprise resource planning: Common myths versus evolving reality. *Business Horizons*, 69–76.

[10] Manetti, J. (2001). How technology is transforming manufacturing. *Production and Inventory Management Journal*, 54–64.

APICS definition for ERP as a method for effective planning and control of all resources needed to take, make, ship, and account for customer orders. There is some truth to this view, but ERP systems are even more comprehensive than simply extensions of manufacturing operations. ERP systems are found in practically all types of large organizations, to include chemical facilities and even universities. MRP-II functions are covered by the PP modules as well as other modules. Figure 2.5 displays the systems typically linked by an MRP system.

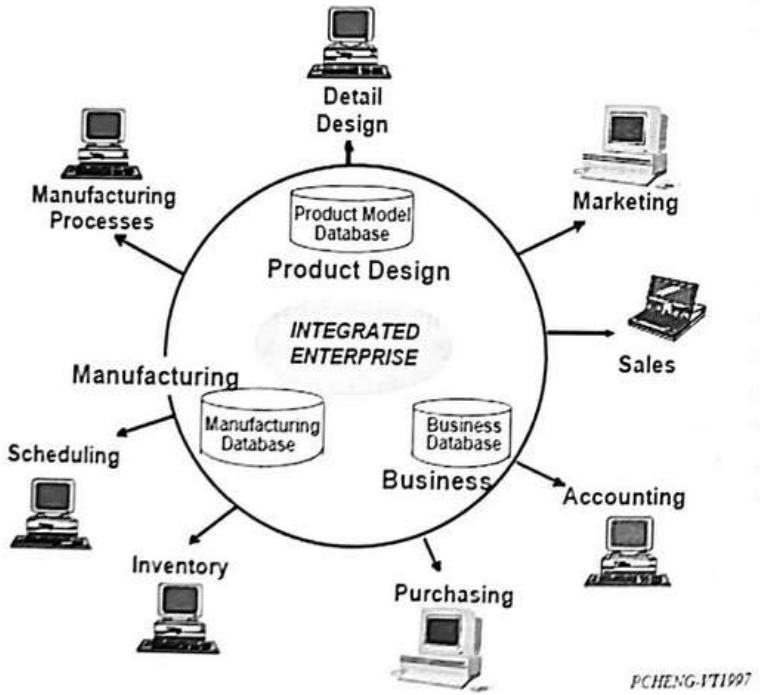


Figure 2.5: MRP System Links.

Source: Adapted from the thesis "Effective Use of MRP-Type Computer Systems to Support Manufacturing," W. Cheng,[11] Virginia Polytechnic Institute and State University.

[11] Adapted from the thesis "Effective Use of MRP-Type Computer Systems to Support Manufacturing," by W. Cheng, Virginia Polytechnic Institute and State University.

2.7 ERP AND ERP MODULES

The launch of ERP systems have spawned a multi-billion dollar global supplier and consulting industry.[12,13] As more and more established organizations realize that they need to form alliances with their customers, partners and suppliers over the internet, ERP systems become a critical global issue.[14] Enterprise Resource Planning is derived from Material Resource Planning.[15] ERP systems are similar to management information systems that incorporate and automate many of the business practices associated with the operations or production aspects of a company. The term ERP was used originally to illustrate a new kind of manufacturing system that has material resource planning, human resource, finance and many other modules on a single database.[16] These typically comprise manufacturing, logistics, distribution, inventory, shipping, invoicing, and accounting. ERP software can lend a hand in the control of many business activities, like sales, delivery, billing, and production; inventory management, and human resource management systems. ERP systems are information systems that "integrate information and information-based processes within and across functional areas in an organization."[17]

They are repeatedly called back office systems indicating that customers and the general public are not directly involved. This is contrasted

[12] Gosain, S. (2004). Enterprise information systems as objects and carriers of institutional forces: The new iron cage? *Journal of the Association for Information Systems*, 5(4), 151–182.

[13] Umble, E. J., Haft, R. R. and Umble, M. M. (2003). Enterprise resource planning: implementation procedures and critical success factors. *European Journal of Operational Research*, 146, 241–257.

[14] Markus, M. L. C. and Tanis, C. (2000). Multisite ERP implementations. *Communications of the ACM*, 43(40), 42–46.

[15] Kesharwani, S. (2005). *ERP System Application Experience and Upsurge*. India, Pragai Prakashan.

[16] Miller, B. D. (2003). What is ERP? www2.cio.com/analyst/report2003.html

[17] Kumar, K. and Van Hillegersberg, J. (2000). ERP experiences and evolution. *Communications of the ACM*, 43(4), 23–26.

with front office systems like customer relationship management systems that deal directly with the customer. ERP systems are "commercial software packages that facilitate the integration of transaction-oriented data and business processes throughout an organization."[18] Thus their intention is to endow with a single integrated data model for several core business functions such as sales, distribution, human resources, accounting and inventory.[19]

So all functional departments that are involved in the operations or in the process of production are integrated in one system. In addition to manufacturing, warehousing, and shipping, this would embrace accounting, human resources, marketing, and strategic management. In the early days of business computing, companies used to write their own software to organize their business processes. This is an expensive approach. Since many of these processes are common across various types of businesses, common reusable software may provide cost-effective alternatives to custom software. Thus ERP software caters to a wide range of industries from service sectors like software vendors and hospitals to manufacturing industries and even to government departments. ERP systems provide cross-organization integration through embedded business processes and are generally composed of several modules including human resources, sales, finance, etc.[20] The greatest benefit that an ERP system delivers to an enterprise is its integration of many of the information systems into one enterprise-wide system.[21] Thus ERP-type systems emphasize a seamless integration

[18] Markus, M. L., Axline, S. *et al.* (2001). *Learning From Adopters' Experiences with ERP: Problems Encountered and Success Achieved. Enterprise systems: ERP implementation and effectiveness.*

[19] Al-Mashari, M. and Zairi, M. (2000). The effective application of SAP/R3: A proposed model of best practice. *Logistics Information Management*, 13(3), 1–10.

[20] Esteves, J. and Pastor, J. (2001). *Enterprise Resource Planning Systems Research: An Annotated Bibliography. Communications of the association for information systems.*

[21] Emily (2004). Integrating ABC & ERP system. www.focusmag.com/back_issues/issue_02/pages/bci.htm.

of data throughout an organization.[22] An enterprise resource planning (ERP) system is a business management system that comprises integrated sets of comprehensive software, which can be used, when successfully implemented, to manage and integrate all the business functions within an organisation. These sets usually include a set of mature business applications and tools for financial and cost accounting, sales and distribution, materials management, human resource, production planning and computer integrated manufacturing, supply chain, and customer information.[23,24]

ERP systems conceptually cover all computing for an organization. The idea is to centralize data and computation, so that data can be entered once in a clean form, and then be used by everyone in the organization (even by supply chain partners outside the organization) with confidence that the data are correct. However, in practice, ERP vendors sell their software in modules. Table 2.2 gives a list of SAP modules in July 2009. This information was extracted from www.oracle.com and www.microsoft.com. As with any Web site, content is subject to change.

SAP has moved away from focusing on modules, using the term solutions that relate to function. These solutions were last seen as:

- Financials.
- Human resources.
- Customer relationship management (CRM).
- Supplier relationship management.
- Product lifecycle management.
- Supply chain management.
- Business intelligence.

[22] Fan, M. Stallaert, J. and Whinston, A. (2000). The adoption and design methodologies of component-based enterprise systems. *European Journal of Information Systems*, 9, 25–35.

[23] Chan, R. (2001). Knowledge management in implementing ERP for SMEs. www.fit.qut.edu.au/student/?n2227169/paper.html.

[24] Yen, D. C., Chou, D. C. and Chang, J. (2002). A synergic analysis for Web-based enterprise resources-planning systems. *Computer Standards & Interfaces*, 24(4), 337–346.

Table 2.2: SAP Modules.

SAP	Description	Oracle
FI	Financial Accounting: general ledger account transaction, generates financial statements.	AP–Oracle payables. AR–Oracle receivables. AX–Global accounting engine. CE–Oracle cash management. FV–Oracle Federal financials. GL–Oracle general ledger.
CO	Controlling: internal management, cost analysis by cost center.	XTR–Oracle treasury.
AM	Asset Management: fixed-asset purchase & depreciation.	CSP–Oracle spares management. EAM–Oracle enterprise asset management.
PS	Project System: manage projects.	PA–Oracle projects.
HR	Human Resources: recruiting, hiring, training, payroll, benefits.	BEN–Oracle advanced benefits. GHR–Oracle Federal HR. HXC–Oracle Time and Labor. PER–Oracle human resources.
PM	Plant Maintenance: preventive maintenance, resource management.	OFA–Oracle assets.
MM	Materials Management: purchasing & raw materials inventory, WIP, finished goods.	BOM–Oracle bills of material. FLM–Oracle flow manufacturing. GMx–Oracle Process Manufacturing. MRP–Oracle master scheduling/MRP.
QM	Quality Management: product inspections, material certifications, quality control.	QA–Oracle quality.
PP	Production Planning: production planning & scheduling, actual production.	CHV–Oracle supplier scheduling.
SD	Sales and Distribution: records sales orders & scheduled deliveries, customer information.	AS–Oracle sales. ASO–Oracle order capture. AST–TeleSales.
CA	Cross application: workflow, business information warehouse, e-mail, industry solutions, CRM, SRM, others.	ASG–Oracle CRM gateway. BIC–Customer intelligence. SIL–Sales intelligence. BIM–Marketing intelligence. BIX–Call center intelligence. CLN–Supply chain trading.

Source: Vendor Web sites.

Only selected Oracle application modules are given here from well over 100 that they offer. Microsoft Dynamics represents the new focus which is part of the evolution in SAP and Oracle systems. CRM, supply chain management, and business intelligence used to be major add-on applications, tapping external software vendors such as Siebel Systems for CRM and SAS, IBM, or SPSS for data mining (business intelligence), and linked to ERP systems through middleware. Oracle purchased Siebel Systems and made their CRM product a module in their system. SAP responded by developing their own CRM. Both vendors provide internal modules for what had been external add-on (bolt-on) products. This demonstrates the continuing evolution of ERP/EIS systems.

2.7.1 Relative Module Use

The degree of module use is reported in Table 2.3 while Figure 2.6 presents information extracted from the three studies of ERP use that we have been monitoring.

The most popular module in the U.S. was financial and accounting, which is the most obvious application needed by an organization. The Swedish study indicated that materials management, production planning, order entry, and purchasing modules were just as popular. In the Korean study, all modules had relatively high usage, but the first five listed were very high at over 90 percent. This might indicate a trend for manufacturing organizations to use more modules, but it might also be specific to Korea. Other modules, given at the bottom of Table 4 and each with adoption rates of less than 50 percent, are either not considered as critical or involve less specificity in best practices. These are similar for both studies, although human resources modules were slightly more popular in Sweden. There have been noted differences in the ease with which different modules are implemented. All financial modules tend to be relatively easy to implement. Those modules relating to manufacturing and human resources also have been implemented with noted success. On the other hand, modules supporting less structured activities, such as sales and marketing, have encountered notable implementation difficulty. Therefore, one reason to implement

Table 2.3: Relative ERP Module Use.

Module	Use — Midwestern U.S. (%)	Use — Sweden (%)	Use — Korea (%)
Financial & Accounting	91.5	87.3	92.5
Materials Management	89.2	91.8	94.1
Production Planning	88.5	90.5	91.5
Order Entry	87.7	92.4	90.5
Purchasing	86.9	93.0	93.1
Financial Control	81.5	82.3	85.0
Distribution/Logistics	75.4	84.8	85.9
Asset Management	57.7	63.3	81.4
Quality Management	44.6	47.5	77.6
Personnel/Human Resources	44.6	57.6	78.4
Maintenance	40.8	44.3	72.2
R&D Management	30.8	34.2	69.5

Source: Mabert *et al.* (2000), Olhager and Selldin (2003), Katerattanakul *et al.* (2006).

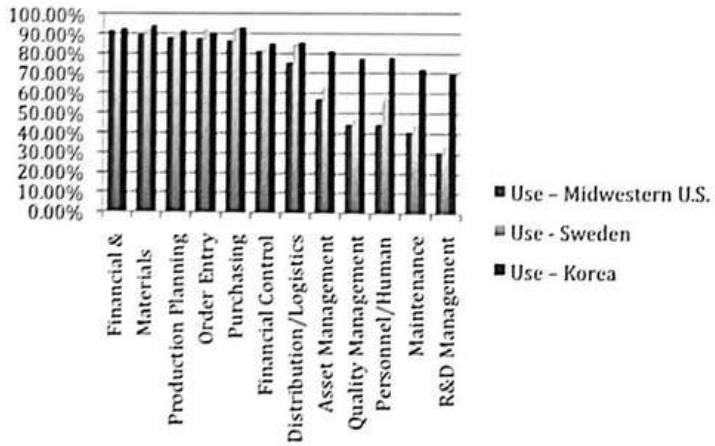


Figure 2.6: Display of Relative ERP Module Use.

ERP in modules is because of the relative need for components of the overall system.

Another (and probably the compelling) reason is cost. Full ERP systems cost a reported $5 million for very small versions to over $100 million

for very large implementations. The fewer modules implemented, the lower the cost. Additionally, it sometimes makes sense to implement the system in bits (phased implementation) rather than try to bring the entire massive system on-line at one time (big-bang implementation). Thus, rolling out an ERP by module sometimes makes sense as well. For a number of reasons, ERP in practice is usually implemented by module.

Often firms will apply the concept of "best-of-breed," mixing modules from different vendors. The Mabert *et al.* study found that a single ERP package was utilized as designed by the vendor in only 40 percent (56 percent in Sweden) of the over 400 respondents to their survey. The most common strategic approach in the U.S. (50 percent, as opposed to 30 percent in Sweden) was to supplement a single ERP package. In fewer cases, the idea of best-of-breed was applied (4 percent in both studies). As might be expected by the enormity of the undertaking, few of the surveyed implementations were entirely constructed in-house (less than 1 percent in the U.S., 2 percent in Sweden).

The idea of best-of-breed approaches is to take advantages of what is perceived as specific vendor-relative advantage in particular areas of application. One vendor's human resource module might be used in conjunction with another vendor's financial and accounting system, and yet a third vendor's materials management modules. Stevens reported one instance of best-of-breed integration triggered by merger.[25] In 1999, Honeywell and AlliedSignal were merged, and the best approaches of each firm's existing ERP systems were examined, with those components judged to be superior retained in the merged firm. Quite often third-party software designed to integrate software applications from several vendors (**middleware** — see Chapter 10) is needed.[26] The role of middleware products is to enable cross-platform operating system communications. This means that software applications such as e-commerce, data warehouses, customer relationship management, supply chain software, and other enhancements can be added to ERP systems.

[25] Stevens, T. (2001). All's fair in integration. *Industry Week*, 24–29.

[26] Bingi, P., Sharma, M. K. and Godla, J. K. (1999). Critical issues affecting an ERP implementation. *Information Systems Management*, 7–14.

Middleware also allows connection of best-of-breed modules to the ERP backbone.

If a firm chooses to utilize their own methods within an ERP, Davenport (1998) gave choices of rewriting the code internally, or using the existing system with interfaces.[27] Both approaches add time and cost to implementation, and thus would dilute the integration benefits of the ERP. The more customization of an ERP, the less ability to communicate seamlessly within system components and across supplier and customer systems.

A related concept is the idea of **federalization**. Davenport (1998) used this term to describe the process of rolling out different versions of an ERP system in each regional unit, tailoring each location's system to accommodate local operating practices.[28] Hewlett-Packard, Monsanto, and Nestle have all used this approach, establishing a common core of ERP modules shared by all units, but allowing other modules to be operated and controlled locally.

2.7.2 Customization Issues

Organizations adopting ERP systems face the problem of what degree of customization to adopt. The best fit with organizational needs would involve intensive reengineering, and development of ERP software by internal staff. But this method is unrealistically slow and expensive. It is much faster and less troublesome to adopt a vendor's software directly. However, in practice, almost every organization adopting an ERP must modify vendor software to at least some degree. Customization is a major issue in ERP design.

2.8 COMPARATIVE COVERAGE BETWEEN MRP, ERP, EIS

Table 2.4 compares characteristics of the three broad periods of software evolution, from MRP through ERP and on to EIS.

[27] Davenport, T. H. (1998). Putting the enterprise into the enterprise system. *Harvard Business Review*, 121–131.
[28] *Ibid.*

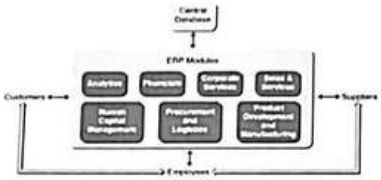
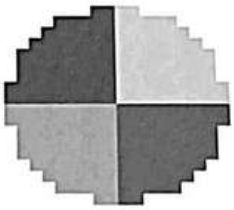
Table 2.4: Comparison of three periods of software evolution.

Particular	MRP	ERP	Extended ERP or EIS
Year	1960–70 MRP-I 1980 MRP-II	1990	2000s
1. Features	MRP-II extends MRP systems to share information with other functional areas. Key component of MRP-II is storing operational information centrally. Just-in-time (JIT) is a Japanese concept which has transformed manufacturing management. It has reduced product cost, and enhanced productivity, product quality and delivery time by as much as 80 percent in some companies. Manufacturers are anxious to implement this system but it needs a high level of co-operation and support from the vendors.	ERP systems seek to integrate all business activities and processes throughout the organization. Goal is to provide real-time information to all employees that need it.	An EIS is a particular category of DSS designed to maintain decision-making at the pinnacle of an organization.

(*Continued*)

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Table 2.4: (*Continued*)

Particular	MRP	ERP	Extended ERP or EIS
2. Diagram	 *Source*: ScmEdge, 2008.	 *Source*: John Wiley, 2006.	 *Source*: EIS Directory service.
3. No parameter for Implementation	Big-Bang Implementation.	Phased implementation, Big-Bang, vanilla Approach.	The Big-Bang approach: organizations implement the new ERP system all at once and scrap their existing legacy systems. The "United Federation" approach: business units/divisions are free to implement independent systems but common processes such as financial reporting are linked across the enterprise. The "Test the Waters" approach: focus is on a few key processes.

(*Continued*)

Table 2.4: (*Continued*)

Particular	MRP	ERP	Extended ERP or EIS
4. Vendors	Conventional Tailor-made Legacy System developed In-house.	SAP-AG, BaaN, Oracle.	Microsoft Dynamics.
5. Time and Cost	Much Less-cost and Less time-consuming	Costs of implementing an ERP are driven by a number of factors including: Number of employees, number of modules that will be implemented, organization's processes must be modified to conform with the ERP system, amount of consulting.	Cost is very high as there is often Web linkage. But in the long-term there is a great potential for cost reduction.
6. Components	Manufacturing and Production related modules.	Includes all the functions of management planning, coordination, decision making, staffing, etc.	Early EIS products were developed for use on high-powered computers, but current products target the client/server platform. These more flexible platforms can adapt to changes in the organization and in technology. Use of real-time data leads to faster, more informed decisions.

Table 2.5: Gartner View of Evolution.

	1980s	1990s	2000s
Competitive focus	Manufacturing	Enterprise	Supply Chain
Enterprise strategy	Inventory reduction	Business process efficiency	Revenue enhancement
Technology focus	Automation	Integration	Inter-operability
Process focus	Departmental alignment	Enterprise-wide closed loop	Customer differentiating
Organizational focus	Department	Business process	Product/Market channel

Note: The term EIS was first coined at MIT in the 1970s. The first EISs were developed by huge firms willing to take risks to increase competitive advantage. By the mid-1980s several vendors had urbanized wide customer bases and EIS technology continues to be shored up today.

2.9 EIS AS PER THE GARTNER VIEW

An EIS is a particular category of DSS designed to maintain decision-making at the top of an organization. An EIS may facilitate a top-level management function i.e. the Managing Director gets an accurate macro-picture of operations, and a synopsis of what competitors are doing. These systems are by and large trouble-free in effortlessly accessing information in ways that can be rapidly absorbed via graphs, charts, etc. Table 2.5 gives this comparison.

2.10 CONCLUSION

Enterprise Resource Planning (ERP systems) has been adopted by many businesses in the past decade. These systems have revolutionized ways of doing business by integrating an organization's business processes, sharing common data and practices across the entire enterprise, producing and accessing information in a real-time environment. Organizations have realized important tangible and intangible benefits from ERP/EIS including general and administrative, expense reduction, margin

enhancement, revenue growth, improved customer satisfaction, personnel and information technology (IT). Despite these benefits, many companies have had serious problems implementing ERP/EIS systems. These problems have caused many companies to abandon their ERP/EIS initiative or implement the system in a limited capacity. Companies have misled ERP/EIS implementation when they see ERP/EIS as a magic way to become competitive, and not as a tool that depends on the way the company uses it. The tool can help companies to become competitive, or it can also take them out of business. A loss of control can occur for many reasons when an ERP/EIS system is implemented; inadequate project definition, planning and implementation is one of the most significant causes for loss of control.

The core idea of ERP/EIS is the complete integration of an organization's computing system. Despite obvious advantages to vendors of each adopting organization installing the entire suite of modules offered, only about half of the implementations seem to be of this nature. Organizations select common modules because not every organization needs every module vendors develop. In fact, vendors seem to recognize this through their recent emphasis on products tailored to specific industries.

There are also other important reasons for implementation of ERP/EIS products different from vendors' original design. A very important one is that full system implementation is very expensive. By selecting particular modules, organizations can cut initial implementation costs significantly. While vendors might argue that in the long run this might be ineffective than full implementation at the beginning, in practice information systems projects rarely go as planned, nor do they tend to stay within originally planned budgets. Thus there is a great reduction in risk to organizations by trying particular modules first, often seeing how the new system is digested by the organization before taking on additional modules.

There is also a difference in the difficulty of implementing different modules. Financial and accounting modules are typically installed first, as they involve the most structured application. This makes it easier to implement, and easier for the organization to digest. Other modules such as materials management and planning also tend to work well. Conversely,

support to less structured environments, such as sales and marketing, tend to be more problematic.

Related to the idea of implementing ERPs in modules are the concepts of best-of-breed, middleware, customization, and federalization. Best-of-breed implementation is idealistic, seeking the best module across vendors and combining whatever mix is viewed as best for the particular organization. This approach is not widely adopted, probably because it involves obvious risks of coordination. It also tends to be more expensive. Middleware is an important type of software making it possible to add on specialty software products. Customization is an approach to implement a vendor's ERP/EIS system tailored to organizational needs. Customization is done internally, as opposed to middleware, which is usually accomplished with third party vendor software. Federalization allows different parts of the organization to utilize different modules, or possibly different levels of customization.

☆ KEYWORDS

- ☆ **By-product:** A by-product is an additional product resulting from the normal processing of the primary item but whose input is not planned; often found in the process industry.
- ☆ **Co-product:** A co-product is an additional product resulting from the process whose input and output is planned.
- ☆ **FIFO stock rotation:** Items may be sequenced for put away using the First In First Out principle.
- ☆ **LIFO storage:** Items may be sequenced for put away using the Last in First Out principle.
- ☆ **Manufactured:** Those items that are designed and fabricated in-house.
- ☆ **Maximum quantity:** The largest quantity that should be planned for any discrete order.
- ☆ **Minimum quantity:** Specifies the smallest order quantity that will be planned by the system.

- ☆ **MRP control code:** This field specifies whether or not the part is under MRP control when planning replenishment orders.
- ☆ **MRP planning policy code:** Specifies how MRP will plan orders for replenishment of material. Planning methods such as order point, lot for lot, period order point, etc. are usually included.
- ☆ **Multiple quantity:** Specifies that material will be ordered in increments e.g., multiple-pack sizes.
- ☆ **Purchased:** Those items that are bought from outside sources.

☆ TERMINOLOGY REVIEW

a. What is meant by material requirements planning [MRP-1]?
b. Define manufacturing resource planning [MRP-II].
c. Explain the different stages of an integrated system.
d. Elaborate how the material requirements planning (MRP) system uses a database approach to manage the large amount of information involved in the material-requirements estimation process.

☆ COMPETENCY REVIEW

1. **Pragmatic Questions**

   a. How do enterprise-wide information systems facilitate in motivating organizations to enlarge or adopt systems?
   b. How do the various functions of manufacturing management accommodate integrated software?
   c. To design an efficient and effective inventory control system for the manufactured housing (MH) industry, it is necessary to accelerate the flow of information and products across the supply chain with the help of an integrated system. Explain.
   d. Comment on this statement! "*To achieve this goal, the system needs three ingredients: generators of quick information flow, generators of*

quick material flow, and facilitators of both information flow and material flow."

2. Briefly comment on:

   a. The alternative perspectives on data to be included in an enterprise-wide area.
   b. The MRP system can be used to develop better planning strategies and to respond quickly to unexpected changes in demand for the organization.
   c. The roles Enterprise Resource Planning Systems (ERPs) play in an organization and the challenging task of implementing and managing the IS function.
   d. The ERP system investigates the interaction among different business processes.

☆ SUGGESTED READING

Books

Brady, J., Monk, E. and Wagner, B. (2001). *Concepts in Enterprise Resource Planning.* Course Technology.

Curran, T. A. and Ladd, A. (2000). *SAP R/3, Business Blueprint,* 2nd Ed. Prentice Hall.

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Sheikh, K. (2003). *Manufacturing Resource Planning (MRP II): With an Introduction to ERP, SCM, and CRM.* New York: McGraw-Hill Professional.

Wallace, T. F. and Kremzar, M. H. (2001). *ERP: Making it Happen — The Implementers' Guide to Success with Enterprise Resource Planning.* John Wiley & Sons.

Journal Articles

Agrawal, A., Minis, I. and Nagi, R. (2000). Cycle time reduction by improved MRP-based production planning. *International Journal of Production Research,* 38(18), 4823–4841.

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White Papers

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